Auto Cad Commands

Initial / Basic settings commands :-

open software. click on clean screen (right side bottom corner)

Limits : lower left corner = 0,0

upper right corner = 15,9

Grid = 0·25

snap = 0·25

Right click on the grid = settings = Display dotted grid in 2D Model space.

Major line every = 1

Adaptive grid = OFF

Display grid beyond limits = OFF

Z Enter A Enter

Use cyan color for construction line

Line Type = LT = load = file = acad = open

scroll down, select.

= hidden2 (for showing hidden lines)

= centre2 (for showing axis lines in solids)

= phantom2 (for showing cutting / section planes in solids)

1 div/spacing = 0·25

1 Auto CAD unit length = 4 div/spacing OR 5 grid points

Make SNAP, GRID, ORTHO, DYN, QP option ON, remaining option OFF. only for drawing inclined objective / line, make ORTHO option OFF Esc, then press Ctrl + S to save the file.

Co

Commands for orthographic and Isometric Projections.

Line = l

Erase = E (select object - enter)

Trim = tr = enter 2 times

Circle = C

Arc = C (centre) = first point = end point

Move = m (select object - enter - base point, corner or centre point)

Copy = co

Dtext = dt = height = 0·15 = angle = 0

Match properties = ma = select source object = select destination objects

Extend = ex = enter 2 times

Pdmode = 3 , Pdsize = 0·15 , Point = PO

polyline = pl (to draw continues line)

spline = spl (to draw curve)

For isometric projections:

Change grid snap from rectangular to isometric snap

For drawing circle · el - i - specify centre of isocircle

OSNAP : make quandrant, tangent option ON, remaining OFF

Commands for projections of Line, Plane, Solid. Section :-

Pdmode = 3

pdsize = 0·15

Point = PO

Polyline = pl (to draw continues line)

Spline = spl (to draw curve)

Dimstyle = dst : current dimensions style : standard

Symbols and arrows : arrow size = 0·12

Text : text height = 0·12

Fit : ✓ tick always keep text

✓ suppress arrows.

Primary units Precision = 0, Scale factor = 100, ok - set current - close

Polygon : pol - no. of sides - E (edge) - specify 1st, then 2nd point

Rotate : ro - select object - select base point - rotation angle

Mirror = MI (to draw axis symmetric object)

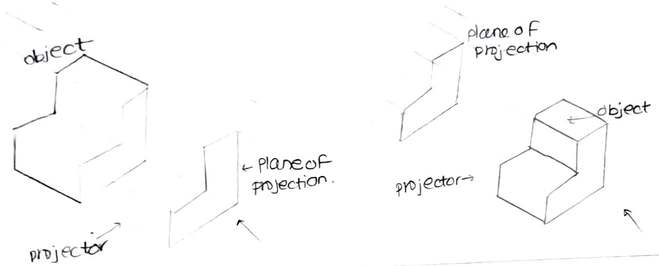
Hatch : for section of solids only

1-Orthographic Projection

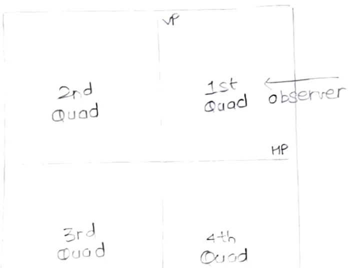
Projections (classification)

- Multi view projection (orthographic projection)
- Single view projection (Pictorial projection)
  - Axonometric projection
    - isometric
    - Dimetric
    - Trimetric
  - Oblique projection
    - Cavalier
    - Cabinet
    - Clinographic
    - General
  - Perspective projection
    - Linear
      - parallel
      - angular
      - oblique
    - Areial

Elements of projections:



Quandrant system



- After the views are obtained on a particular plane
- HP rotated about XY, in clockwise direction to bring it in a plane with the VP
- PP is rotated about X1Y1 away from the object to bring it in a plane with VP

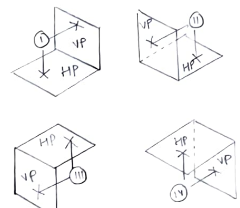
- Intersection of horizontal plane (HP) and vertical plane (HP) and vertical plane (VP), makes four quadrants, measuring in a anticlockwise direction as 1st, 2nd, 3rd 4th quadrant.
- It is also called as 1st, 2nd, 3rd and 4th angle method of projections.
- The positions of an object in space can be determined by these four quandrants, by placing the object in any of these quadrants.

Different reference planes are

- Vertical Plane (VP)
- Horizontal Plane (HP)
- Profile Plane (PP)
- FV is projected on VP
- TV is projected on HP
- SV is projected on PP

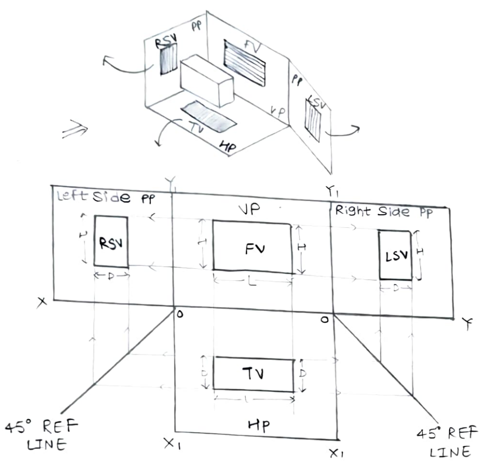
Quadrant	Wordings in the problem Statement	Position of views
1st	above HP and infront of VP	FV above, TV below XY line
2nd	above HP and behind VP	both FV, TV above XY line
3rd	below HP and behind VP	TV above, FV below XY line
4th	below HP and infront of VP	both FV and TV below XY

1st Quadrant 2nd Quadrant

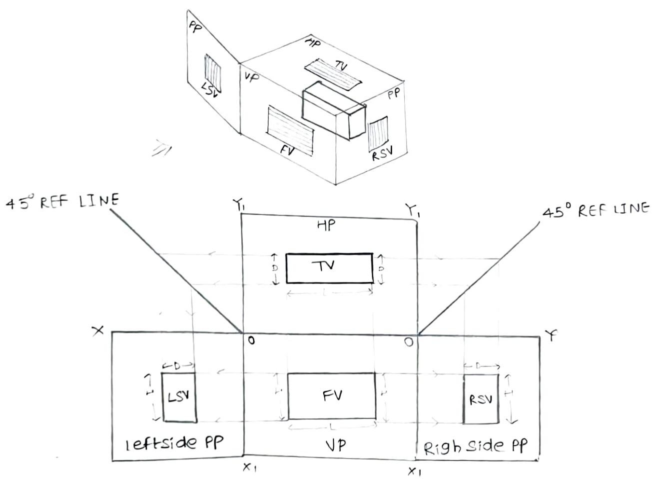


3rd Quadrant 4th Quadrant

... quadrant Method of Projection



3rd angle, 3rd quadrant method of projections

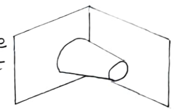


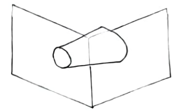
Method of Projection

1st angle	3rd angle
Object is placed above HP and front of the VP	Object is placed below HP and behind the VP.
Object is placed in between observer and plane of projectn	plane of projection is placed in between observer and object
Plane of projection is assumed to non transparent	Plane of projection is assumed to be transparent
When views are drawn in their relative position, TV comes below FV. whereas RHSV is drawn to the left side of FV and vice versa	When views are drawn in their relative position, TV comes above FV. Whereas RHSV is drawn to the right side of FV and vice versa
This system is followed in India and European countries	This System is follow in USA

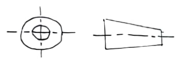
Symbolic Representation of 1st and 3rd Quadrant

Truncated cone in I Quadrant

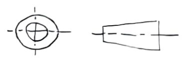




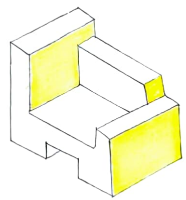
Truncated cone in III Quadrant



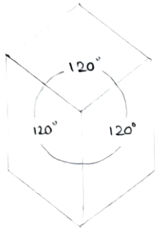
symbolic representation for I angle method of projection



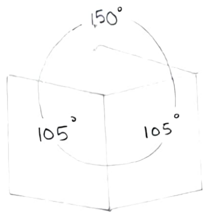
Symbolic representation for III angle method of projection



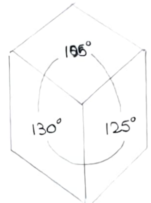
2-Isometric Projetions



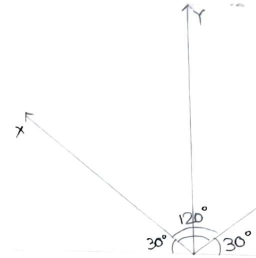
3 Equal Angles
ISOMETRIC

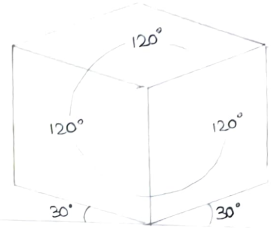


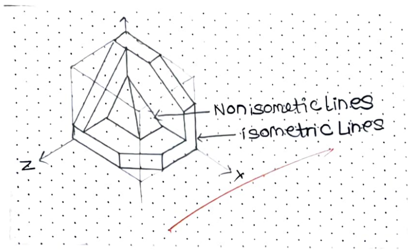
2 Equal Angles
Dimetric



No Equal Angles
Trimetric







Isometric Lines	Non-Isometric Lines
Lines or edges which are parallel to either of these X, Y, Z axis are called Isometric Lines.	Lines or edges which are not parallel to either of these x, y, z axis are called as non-isometric lines
They are also called as straigh Lines	They are also called as Inclined lines
They are showing their true lengths	They are showing their apparent lengths

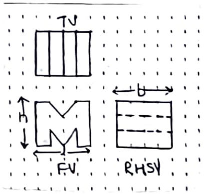
Isometric Axes, Lines and Planes

1) Isometric Axes - The three lines AL, AD, AH, meeting at point A and making 120° angles with each other

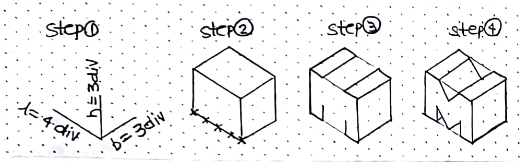
2) Isometric Lines - Lines parallel to Isometric Axes

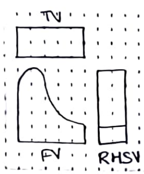
3) Isometric Planes - The planes representing the faces of the cube as well as other planes parallel to these planes are called Isometric Planes.

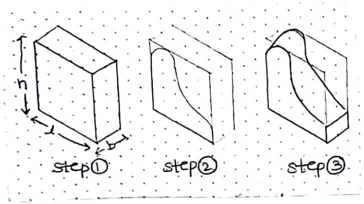
Isometric view	changes to shape
sphere	circle
circle	ellipse
square	rhombus
rectangle	parallelogram



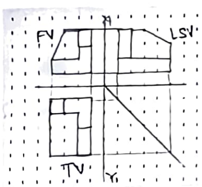
Maximum length = 4 div
height = h = 3 div
breadth = 3 div

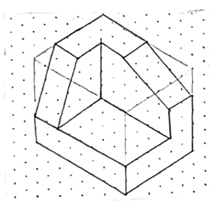


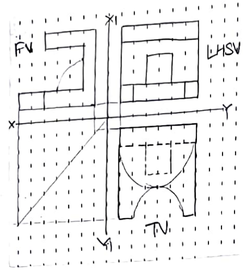


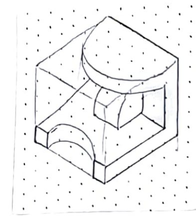


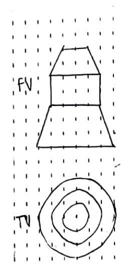
l = 5div
h = 5 div
b = 2div



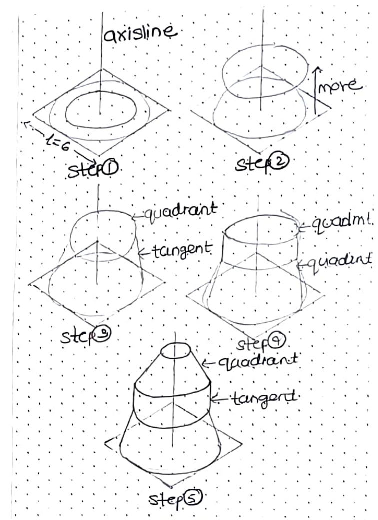






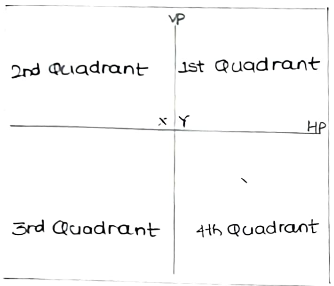


h = 7div
l = 6div



3A-Projection of Point/Lines

Quadrant System



- after the views are obtained on a particular plane:
- HP is rotated about XY, in clockwise direction to bring it in a plane with the VP
- PP is rotated about X1Y1 away from the object to bring it in a plane with the VP

① Intersection of horizontal plane (HP) and vertical plane (VP) makes four quadrants, measuring in a anticlockwise direction as 1st, 2nd, 3rd, 4th quadrant.

② It is also called as 1st, 2nd, 3rd, 4th angle method of projections.

③ The position of an object in space can be determined by these four quadrants, by placing the object in any of these quadrants.

④ Different reference planes are:
Vertical Plane (VP)
Horizontal Plane (HP)
Profile Plane (PP)

FV is projected on VP
TV is projected on HP
SV is projected on PP

Quadr	Wordings in problem statement	Position of views
1st	above HP and infront of VP	FV above and TV below XY line
2nd	above HP and behind VP	both FV and TV above XY line
3rd	below HP and behind VP	TV above and FV below XY line
4th	below HP and infront of VP	both FV and TV below XY line

Object	Point A	Line AB
its FV	a'	a' b'
its TV	a	ab
its SV	a"	a" b"
TL in FV	—	a'b1'
TL in TV	—	ab_1

Key Parameters about Projections of Lines.

- TL (true length) in FV = a'b1'
- TL (true length) in TV = ab1
- Angle of TL with HP = θ
- Angle of TL with VP = ϕ
- Angle of FV with XY = α
- Angle of TV with XY = β
- Length of FV (apparent FV) = a'b'
- Length of TV (apparent TV) = ab
- Position of A = distance of a & a' from XY line
- Position of B = distance of b & b' from XY line
- Distance between End Projectors.

* TL must be same in both FV and TV
* True length > Apparent length
* True Angles ($\theta \phi$) < Apparent angles (α, β)
* Angles θ and α construct with a
* b' and b1' on the same locus
* b and b1 on the same locus.

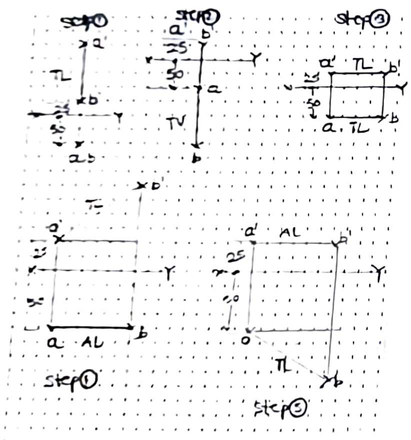
From the given apparent length, rotate (by drawing circle) to make it parallel and then project to get True length
From the given True length, project (by drawing verticle straight line) and then rotate to get the apparent length.

Sample cases of the Line: Questions

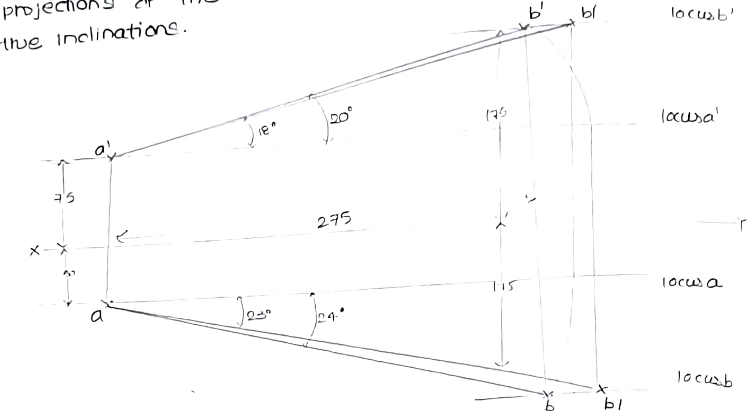
① Line AB Perpendicular to HP and Parallel to VP
② Line AB. Perpendicular to VP and Parallel to HP
③ Line AB. parallel to both HP and VP
④ Line AB inclined to HP and parallel to VP
⑤ Line AB inclined to VP and parallel to HP.

In all above cases. following assumptions are taken:
Line AB=100 mm. lies in First quadrant i.e FV above and TV below XY line. Point A 25 mm above HP and 50mm infront of VP. TL = true length = 100 mm. AL = apparent length

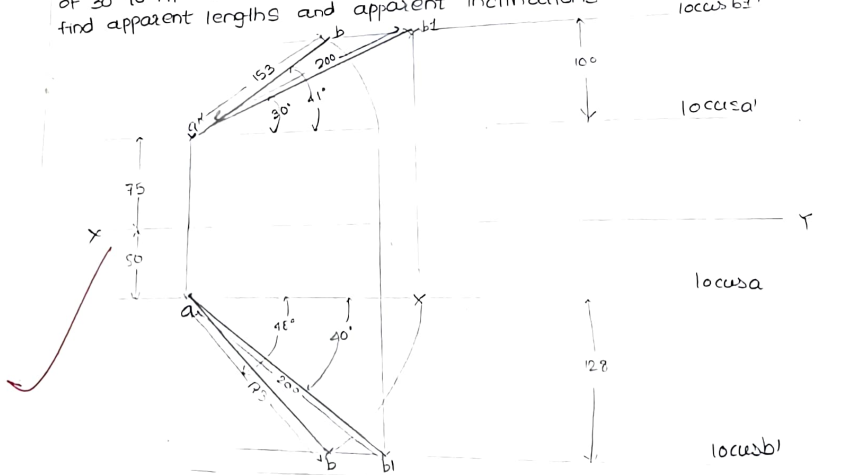
Note: Line cannot be perpendicular to both HP, VP.



① Line AB has its end A 75 mm above HP and 50mm infront of VP. The other end is 175 mm above HP and 175 mm infront of VP. The projector distance is 275 mm. Draw projections of the line AB. obtain the true length and true inclinations.



② Line AB 200 mm long has its end A 75mm above HP and 50 mm in front of VP. Endpoint B is in 1st quadrant. The line makes an angle of 30° to HP and 40° to VP. Draw projections of straight line AB and find apparent lengths and apparent inclinations.



4. Projection of Planes

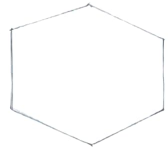
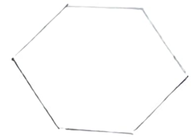
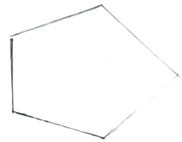
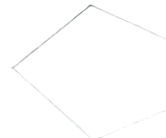
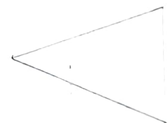
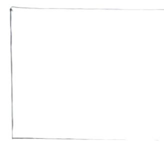
Diff. NO	Line Problems	Plane Problems.
1	Line Problems are asked in all quadrant system	Plane problems are asked only in first quadrant system
2	Lines are obtained by joining the Points	Planes are obtained by joining the lines
3	Lines cannot perpendicular to both HP and VP	Planes cannot be parallel to both HP and VP
4	In Line problem statement, the words are True lengths and Apparent Lengths	In Plane problems statements the words are True shapes and Apparent shapes.

Steps to Solve Projections of Planes:

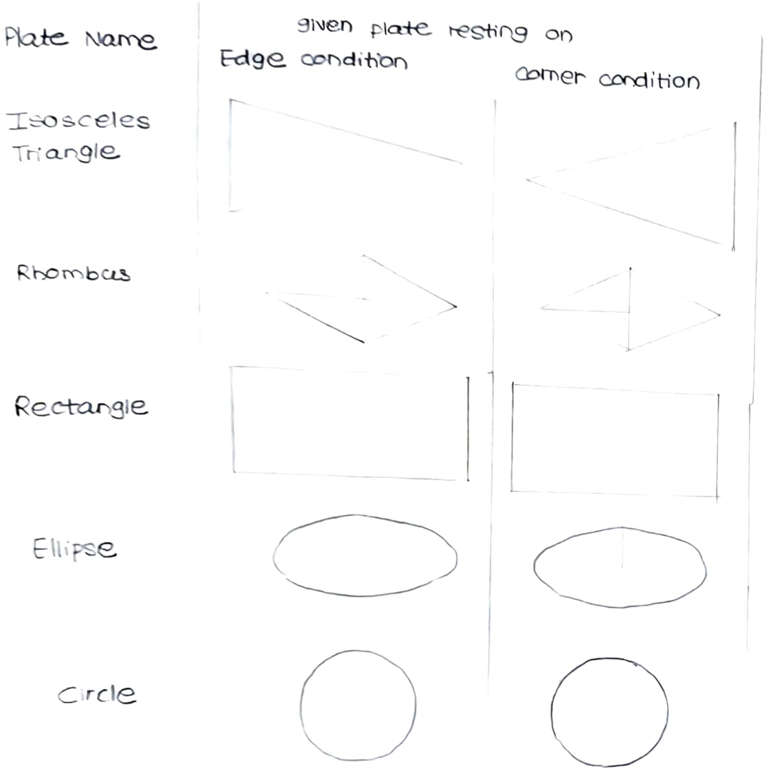
step①:
Make surface of given plate, parallel to HP or VP. With this draw FV and first TV, constituting first step.
a) If a given plate is resting on HP, assume surface parallel to HP. In this case, TV shows true shape of the Plate, whereas FV shows a line/edge view.
b) If a given plate is resting on VP, assume surface parallel to VP. In this case, FV shows true shape of Plate, whereas TV shows a line/edge view.

step②:
Consider surface angle with HP or with VP. With this draw second FV and second TV, constituting second step.
Surface inclination can be given by 3 ways.
a) By providing direct surface angle with HP or with VP.
b) By Apparent shape in FV or in TV
c) By providing opposite edge/corner linear distance from HP/VP

step③:
Cosider Edge/side/diagonal angle with HP or with VP. with this draw Third FV and Third TV, consituting Third step.

	given plate resting on	
Plate Name	Edge Condition	Corner condition
Hexagon		
Pentagon		
Equilateral Triangle		
Square		

Draw all above equal side plane shape by using command POLYGON. Type POL – no. of sides – E (edge) – specify first endpoint of edge.

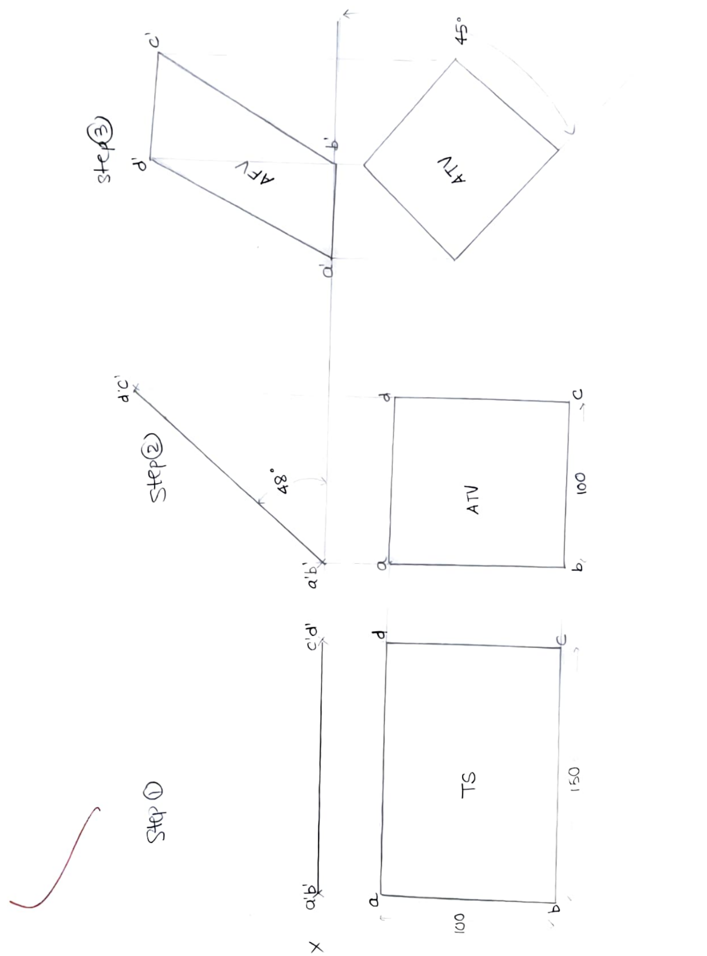


Draw above plane shape by using respective command i.e Line, ellipse or circle.

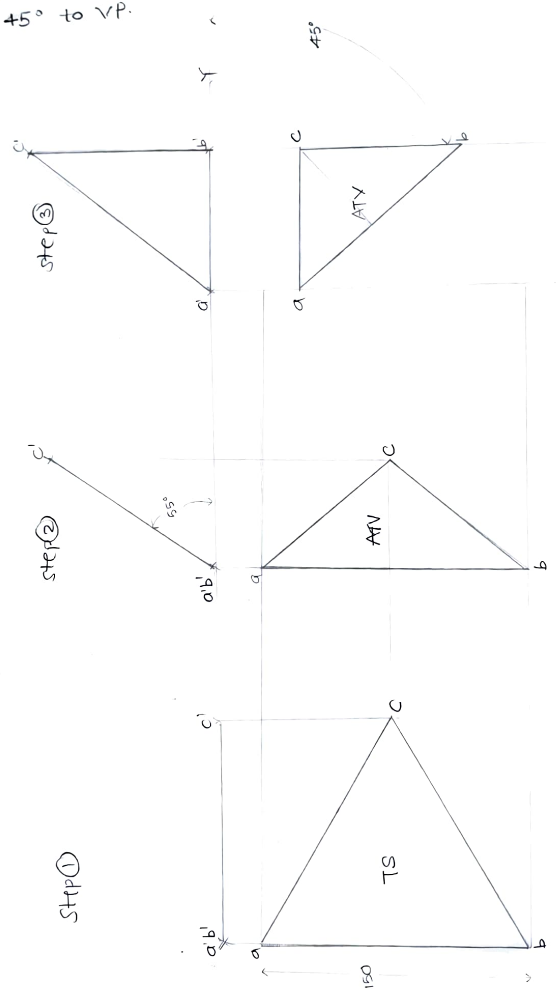
(a) If plane is resting on one of its edge, then draw the true shape of the surface with the resting edge perpendicular to XY line.

(b) If the plane surface is resting on one of its corner, then draw true shape of the plane surface such that the line joining the resting corner and the centre of the plane, parallel to the XY line.

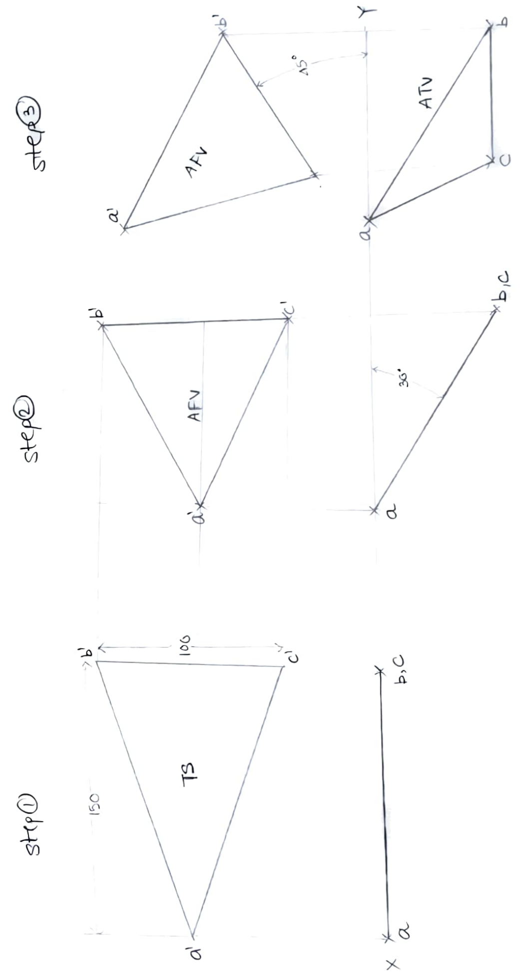
(1) Rectangular Plate of dimension 150 x 100 mm is resting on its smaller edge on HP. Its surface is inclined 48° to HP. Draw its projection, when resting edge make 45° to VP.



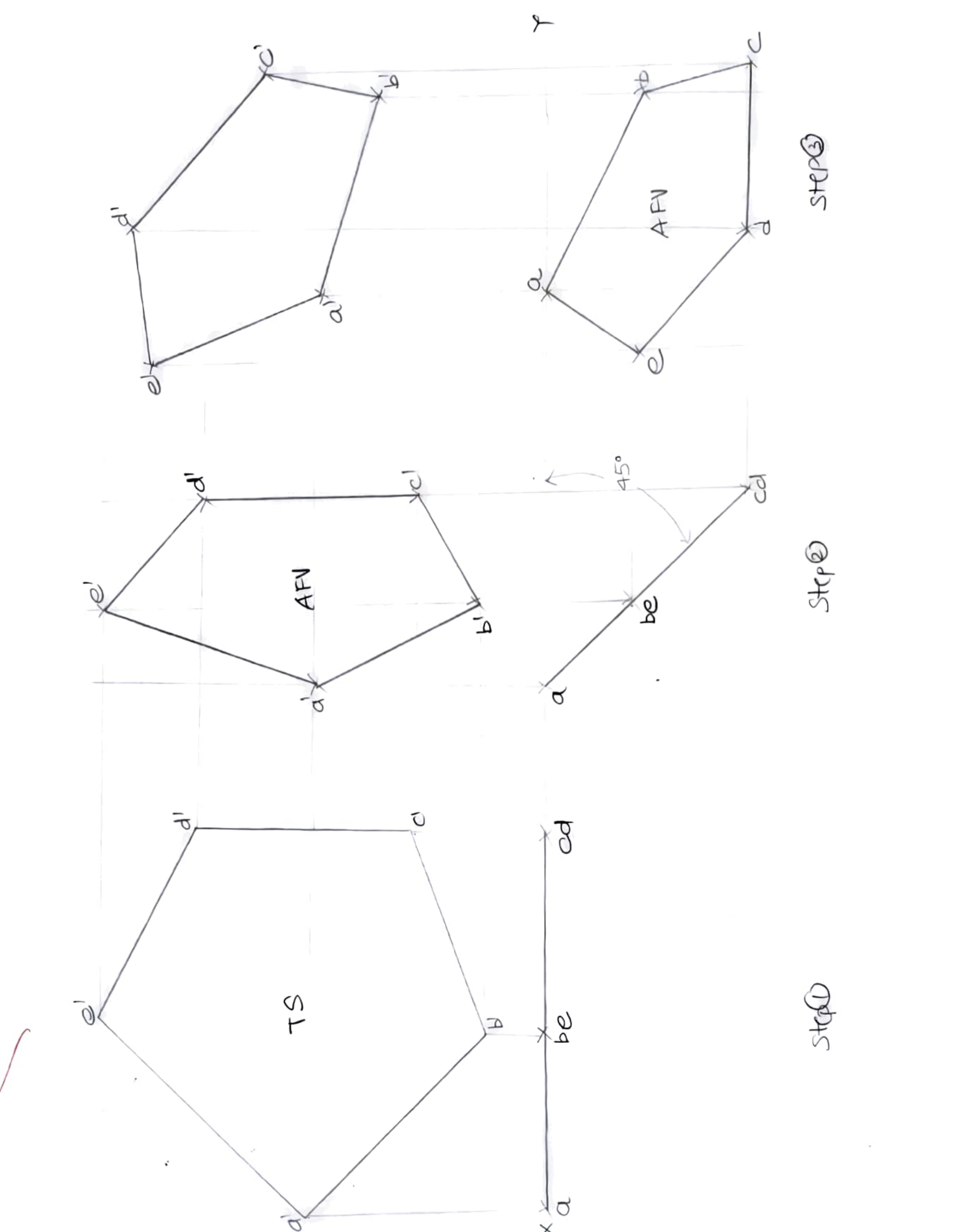
) Equilateral triangular plate of side 150 mm is resting
n its base on HP, with surface is inclined 55° to HP. Draw
s projections, when base of that triangle makes
n angle of 45° to VP.



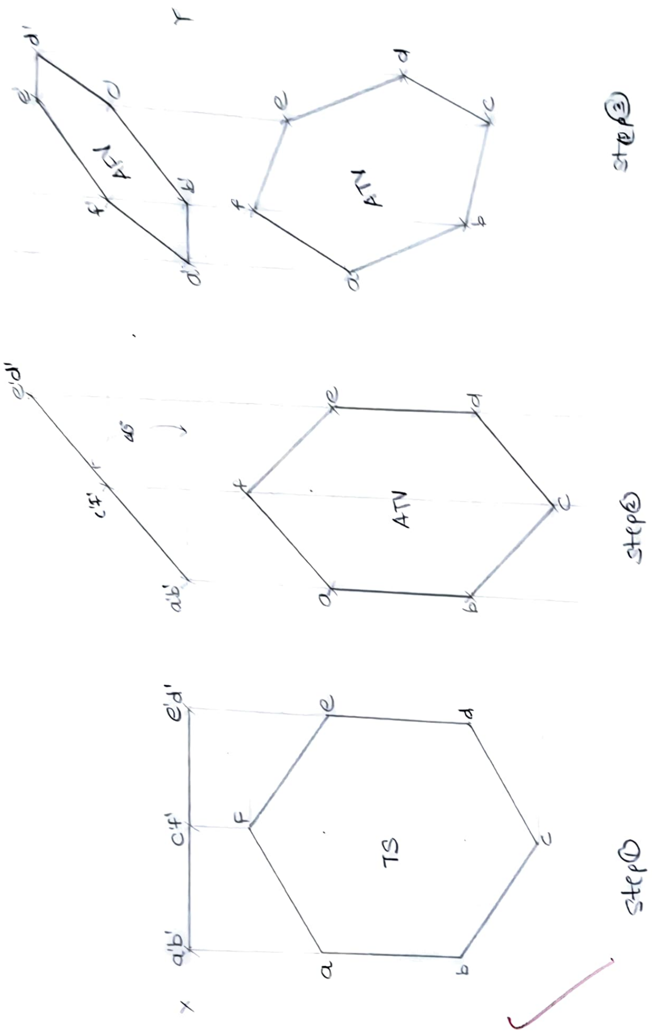
(3) Isosc... corner on VP. Its
altitude 150 mm is resting on its
surface is inclined 30° to VP. Draw its projections, when
base of that triangle makes an angle of 45° to HP.



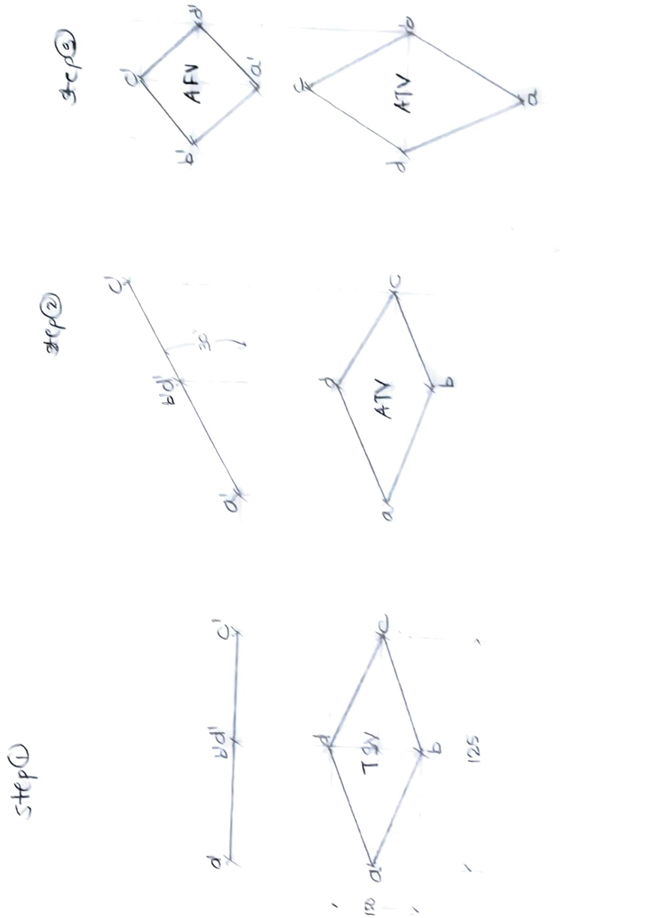
(5) A pentagonal plate of 100mm side is resting on one of its corners on VP. The surface is inclined at 45° with the VP. The edge, opposite to that corner is inclined at 35° with HP. Draw projections of Plate.



(6) Hexagonal Plate of side 75mm is resting on one of its edges on HP. and inclined 60° to VP. Surface of a plate is inclined 40° to HP. Draw its Projections.



(7) Rhombus plate having diagonals 125 mm and 50 mm long. The smaller diagonal of which is parallel to HP a
VP. while other diagonal is inclined 30° to the HP. Draw
its Projections.



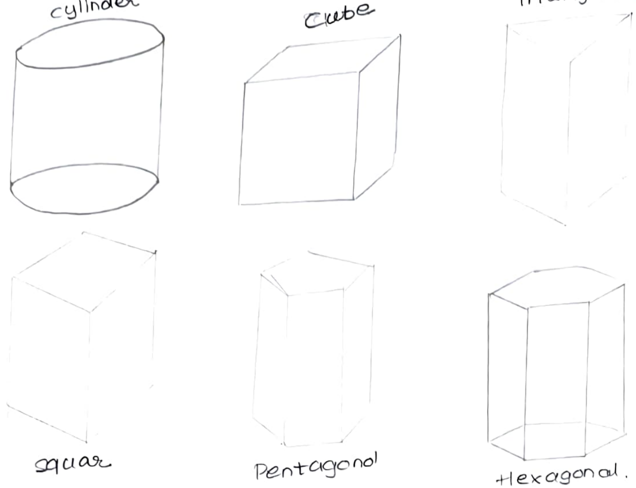
5. Projection of Solids

Diff	Plane problem	Solid problems
1	Planes are obtained by joining the lines.	Solids are obtained by joining the Planes
2	As planes are inclined to both HP and VP, plane problems are of 3 steps	As solids are inclined to one Plane (HP or VP) and parallel to other plane. solid problems are of 2 steps.
3	In the Second step. Consider surface inclination with HP or VP	In the second step, consider axis inclination with HP or VP.
4	In the first step. make surface of a given plate parallel to HP or VP, to which it is resting	In the first step, make axis of a given solid perpendicular to HP or VP, to which it is resting
5	In the Third step, consider side / edge / diagonal inclination.	There is No Third step.

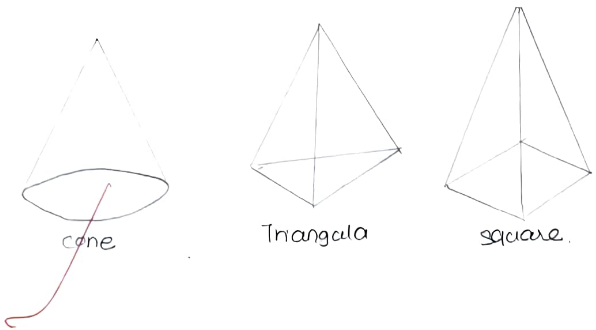
Solids of revolution:
When a solid is generated by revolutions of a plane figure about a fixed line (axis) then such solids are named as solids of revolution.

Classification of solids

Group (A)
Solids having top and base of same shape

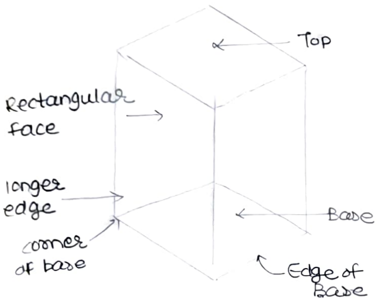


Group (B)
Solids having base of some shapes and just a point as a top, called as apex.

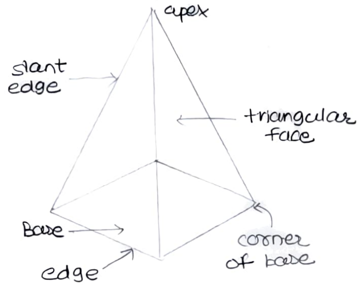


Dimensional Parameters of different solids

Square Prism

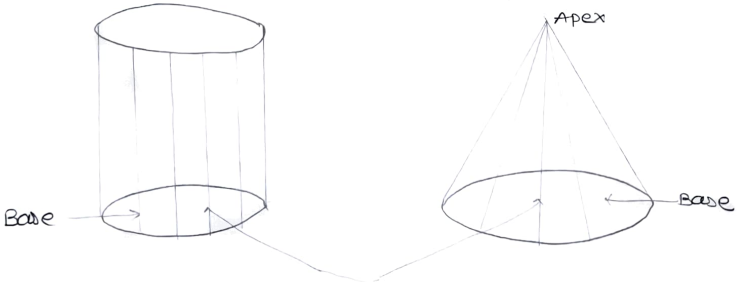


Square Pyramid



Cylinder

Cone



Generators
Imaginary lines generating curves surface of cylinder and cone.

Steps to solve projection of solids

Step ① :

Solids may rest on HP or on VP, on one of its

a) Edge of base
b) Corner of base
c) circumferential base point or curved surface
d) slant / vertical face
e) slant / vertical edge

In all the above cases, assume the solid Standing on HP or VP, to which it is resting. Thereby make Axis of a given solid perpendicular to HP or VP. Draw True shape of the base of a given solid first. With this, draw First FV and First TV, constituting first step.

Step ② :

Consider axis / solid inclination. With this, draw First FV and First FV, constituting second step.
Axis / Solid inclination can be given by 5 ways:

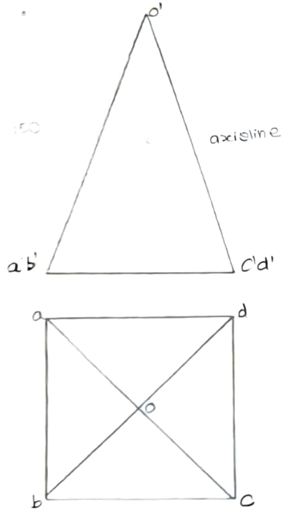
a) By giving direct axis angle with HP or VP
b) By complete base angle with HP or VP
c) By resting on one of its slant / Vertical face, on HP or VP.
d) By resting on one its slant / vertical edge, on HP or VP
e) By providing apex / apposite edge / corner, linear distance from HP or from VP.

① Square pyramid of base edge 100mm and height 150 mm is resting on one of its base edge on HP, such that axis is inclined 45° to HP and parallel to VP. Draw its projections.

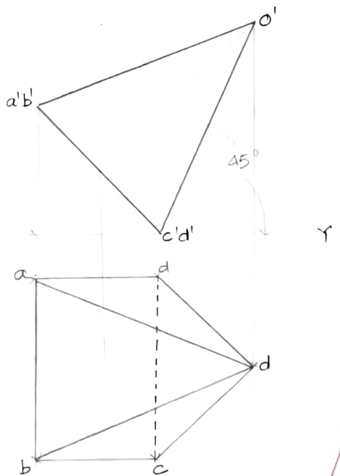
stage①: Draw square plate at edge condition as TS of the base in TV as it resting on its base edge on HP. Obtain centroid "O". join with base edges. Then draw axis line passing from the centroid and draw FV.

stage ②: Rotate axis line at a given axis angle along resting edge a'b' or o'd'. Join all points in TV (2nd TV) by following 3 parameters.

a) Join all boundary points by continuous lines
b) Lines / edges emerging from the farthest point to the observer to be joining by hidden lines
c) Lines / edges emerging from the nearest point to the observer to be joining by continuous lines.



step①

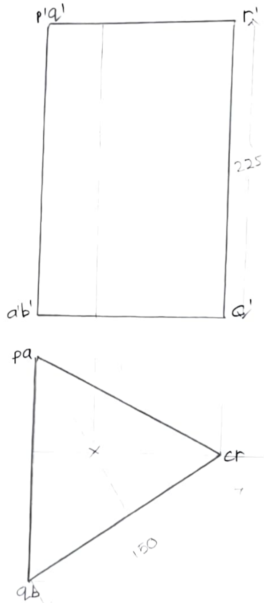


step②

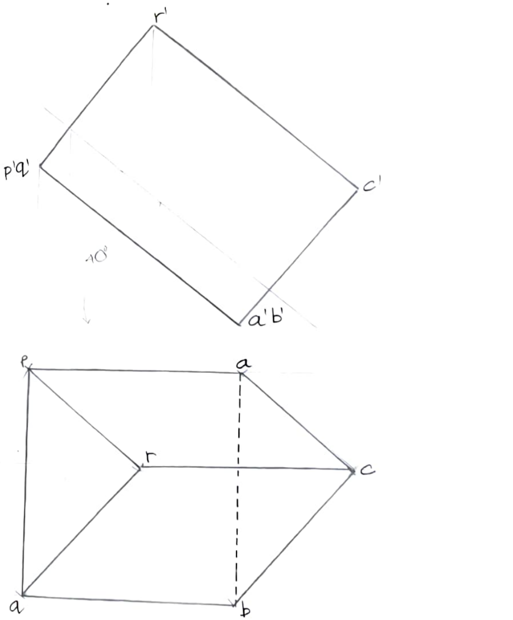
② Triangular prism, base edges 150mm and height 225 mm is resting on one of its base edges on HP with an axis inclined 40° with the HP and parallel to VP. Draw its projections.

stage①: Draw equilateral triangle under edges condition as a TS of the base of a triangular prism in TV and then draw its respective FV.

step②: copy FV at some distance on XY line, then rotate copied FV along resting edge a'b', such that axis inclined 40° to HP. Then join all points in TV by those 3 parameters.



step①

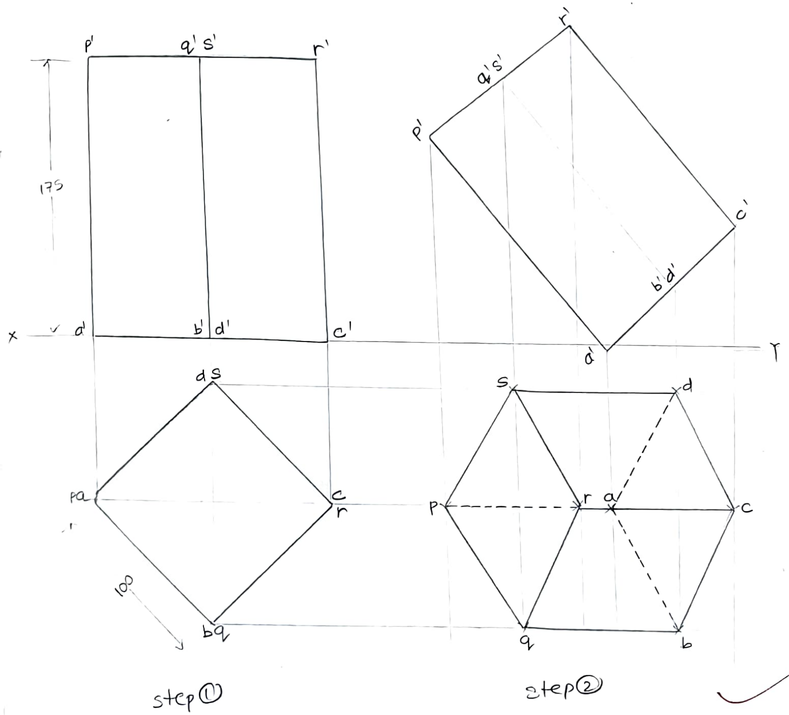


step②

(3) Square prism of base side 100mm and height 175mm is resting on its base corner on HP, such that axis is inclined 40° to HP and axis parallel to VP.

stage① : Draw square plates at corner condition as TS of the base in TV as its resting on its base edge HP.

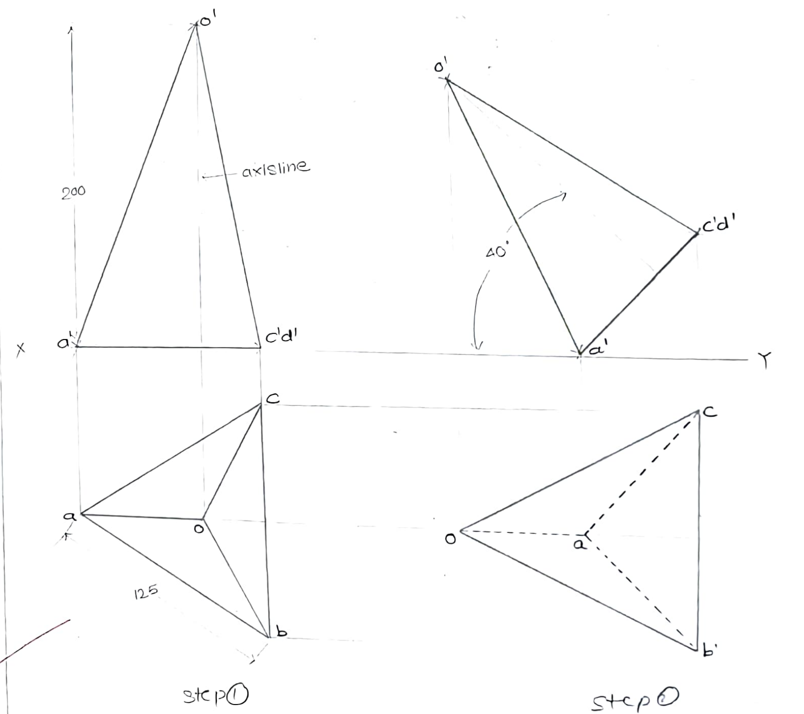
stage②: Cope FV at same distance on XY line, then rotate copied FV along resting corner a' OR c' such that axis inclined 40° to HP. Then join all points in TV by those 3 parameters.



(4) Triangular Pyramid of base side 125 mm and height 200 mm is resting on its base corner on HP, such that the axis is inclined 40° to HP and axis parallel to VP.

stage① : Draw equilateral triangular plate at corner condition as TS of the base in TV, as it resting on its base corner on HP. Obtain centroid "o". Join with base edges. Then, draw axis line passing from centroid and draw FV

stage②: copy FV at same distance on XY line, then rotate copied FV along resting corner's a', such that axis inclined 40° to HP. Then join all points in TV by those 3 parameters.



⑤ Pentagonal Pyramid of base edge 100mm and axis 200 mm is resting on one of its base edge, with the axis is inclined 35° to HP and parallel to VP. Draw its Projections.

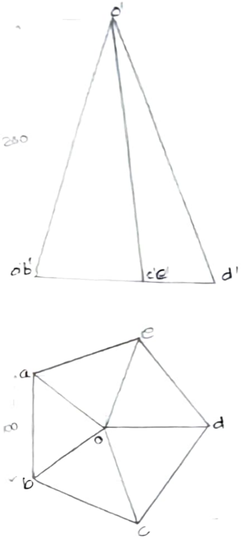
stage①: Draw Pentagonal Plate at edge condition as a True shape of the base in TV. Obtain centroid. from that draw axis line and draw FV.

stage②: Join these points by following 3 parameters

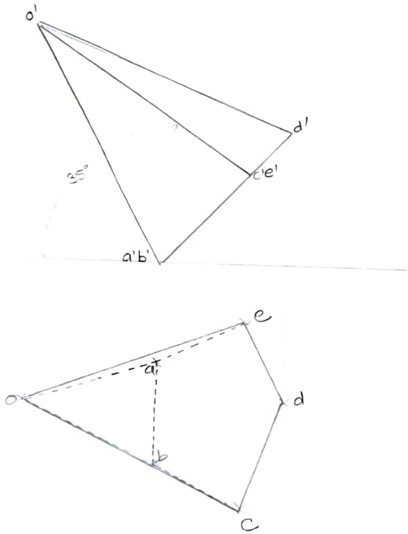
a) Join all boundary points by continuous lines

b) Lines / edges emerging from the farthest point to the observer to be joined by hidden lines.

c) Lines / edges emerging from the nearest point to the observer to be joined by continuous lines.



step①

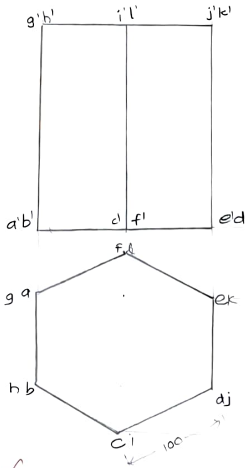


step②

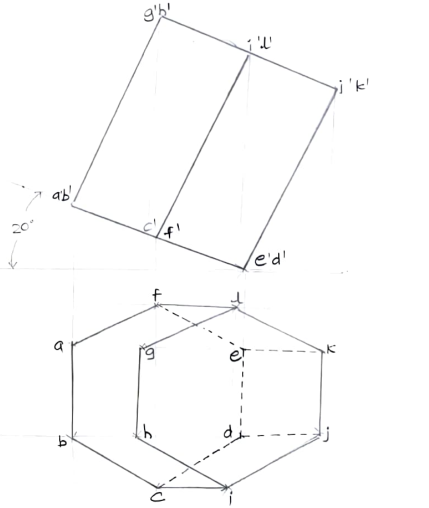
⑥ Hexagonal Prism of base side 100 mm and height 150 mm is resting on one of its edge of base, such that base is inclined 20° to HP and axis parallel to VP. Draw its Projections.

stage ①: Draw hexagonal Plate at edge condition as a The shape of the base of hexagonal prism in TV. As it is prism, so bottom base of a hexagon is given naming as a, b, c, d, e, f, whereas top face of a hexagon is named as g, h, i, j, k, l. Then draw its respective FV.

stage②: Join all points in TV by those 3 parameters.



step①

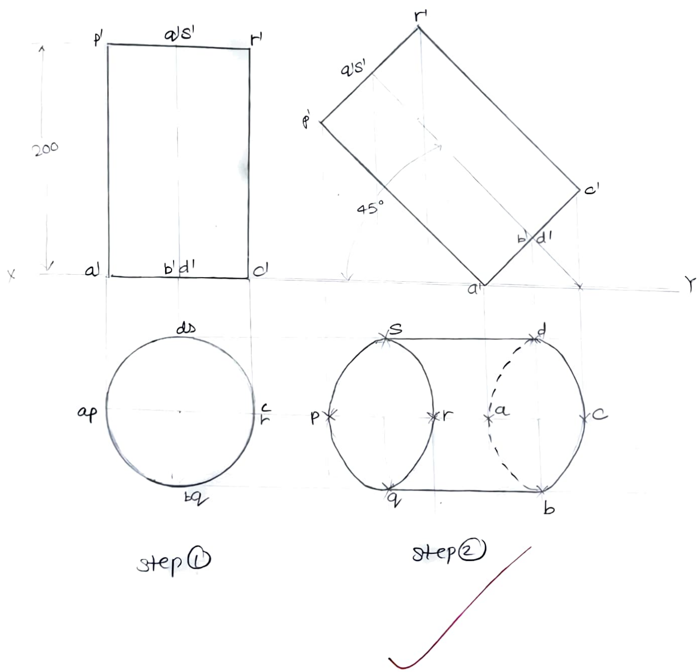


Step②

(7) Cylinder of diameter 150 mm and height 200 mm is resting on its circumferential base point on HP, with an axis inclined at 45° to HP and parallel to VP. Draw its projections.

stage ①: Draw circle as TS of the base of a cylinder in TV and then drawn its respective FV

stage ②: copy FV ato some distance on XY line, then rotate copied FV along circumferential point a' or c', with an axis inclined 45° to HP. Then join all points in TV by those 3 parameters.



6. Section and Development of Solids

Section of Solids

① In engineering practice, it is often required to make the drawing showing the interior details of the object, which are not visible to the observer from outside.

② If the object is simple in its construction, the interior portion of the object can be easily interpreted by the dotted lines in the orthographic projection.

③ But in cases where the interior construction of the object is complicated, it cannot be clearly interpreted by the dotted lines. In such case views can be drawn by cutting the object apart by an imaginary cutting plane and the portion between the cutting plane and the observer is assumed to be removed so as to show the internal constructional detail of the invisible feature. Such views are known as sectional views.

④ The action or (imaginary) process of cutting a solid is called sectioning and the imaginary plane by which the object is assumed to be cut is known as section plane or cutting plane.

Types of Section Planes

Depending upon the positions of the section planes wrt reference plane, there are different position of section

① section plane perpendicular to VP and parallel to HP
② section plane perpendicular to VP and inclined to HP (AIP)
③ section plane perpendicular to HP and parallel to VP
④ section plane perpendicular to HP and inclined to VP (AVP)
⑤ section plane perpendicular to both HP and VP
⑥ section plane inclined to both HP and VP

Note. Out of above mentioned position of cutting/section planes, 1st and 2nd type are in syllabus

Terms involved for solving section of Solids Problems

① Surface produced by cutting the solids by section plane is called section.

② Apparent shape of section: If the cutting plane is inclined to one of the reference plane (HP or VP) and perpendicular to other reference plane, then the projection of section on that reference plane to which cutting plane is inclined will show its apparent shape of the screen.

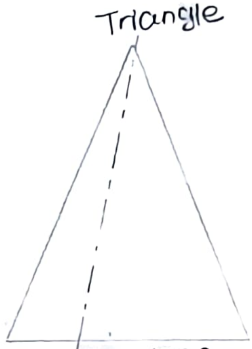
③ True shape of section: If the cutting plane is parallel to one of the reference plane then the projection of section that reference plane will show its true shape of section.

④ In order to view True shape of the section after cutting solid, direction of observer must be perpendicular to section/cutting plane. If it is inclined, then observer can view its apparent shape of the section.

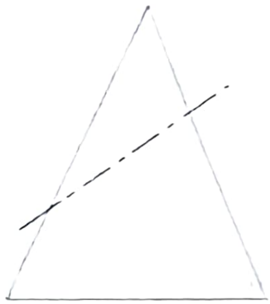
⑤ The cut surface of the object is shown by series of narrow, inclined lines parallel to themselves called section lines or cross-hatching lines.

⑥ The process of drawing sections lines is called hatching.

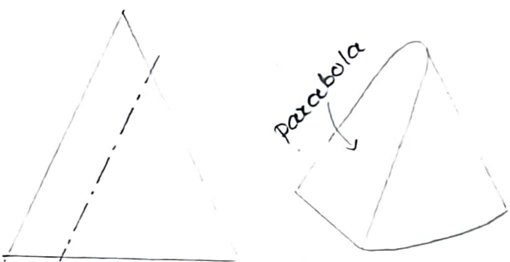
Different positions of section planes and its respective true shape of sections obtained :



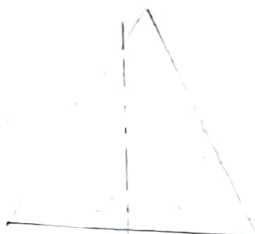
section plane Through Apex of cone



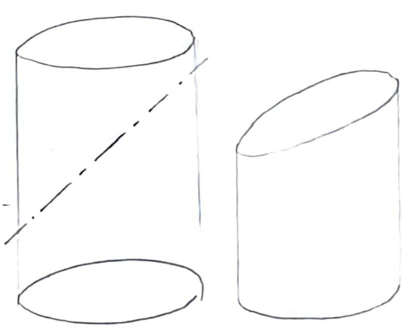
section plane Through Generators of Cone



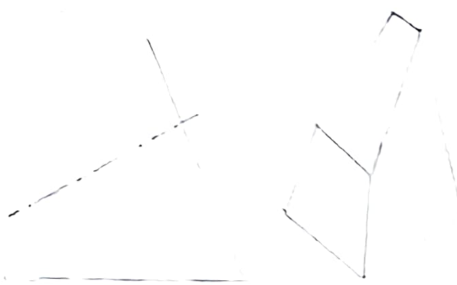
section plane parallel to end generator of cone



section plane Parallel to Axis of cone



setion plane passing through all generator of a cylinder



section plane passing through all slant edges of a sq. pyramid.

Development of surface solids

Definition -

If the solid is assumed to be hollow and made up of thin sheet. It is cut at one side and unfolded all the surfaces of the solid, and then the shape of the unfolded sheet (surface area) is called development of surface of a solid.

Application -

There are so many products or objects which are difficult to manufacture by conventional manufacturing process because of their shapes and sizes. Those are fabricated in sheet metal industries by using development of surfaces of such objects. To make any object out of sheet metal, first prepare its development and then cut the sheet metal according to the shape and size of the development and then roll it or fold it into the required shape.

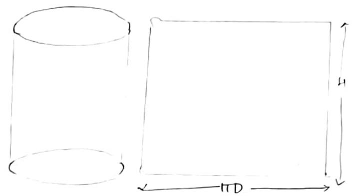
Examples -

Boiler shells and chimney, pressure vessels, trays, boxes, containers, funnels, feeding hoppers, buckets, ducts, large pipe sections, body of automotives, aircrafts, ships, etc

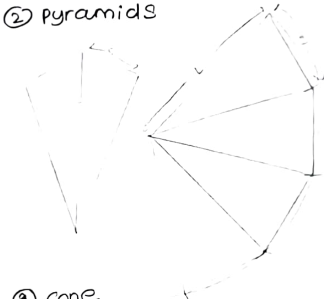
Types of Development of surface of Solid

Lateral Development: It is the development of surfaces of solid, including lateral surface but excluding top and bottom surface:

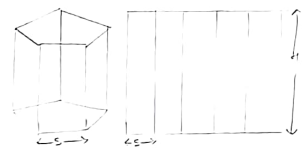
① Cylinder: A Rectangle



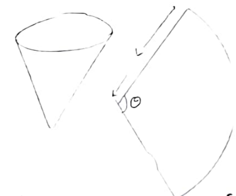
② Pyramids



③ Prisms: No. of Rectangles.



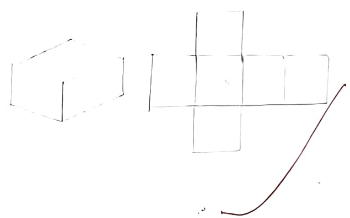
④ Cone



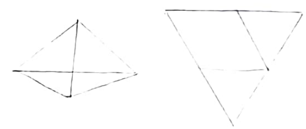
R = Base radius
L = slant height
$\theta = \frac{R}{L} \times 360°$

Full Development: It is the development of surface solid, including lateral, top and bottom surface.

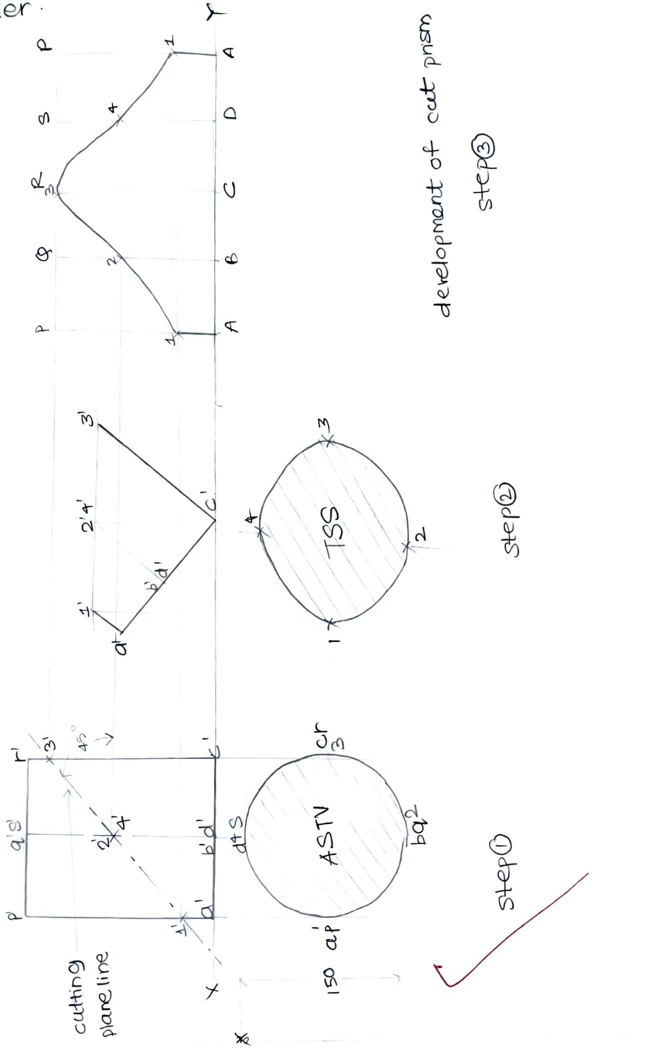
① Cube: six square



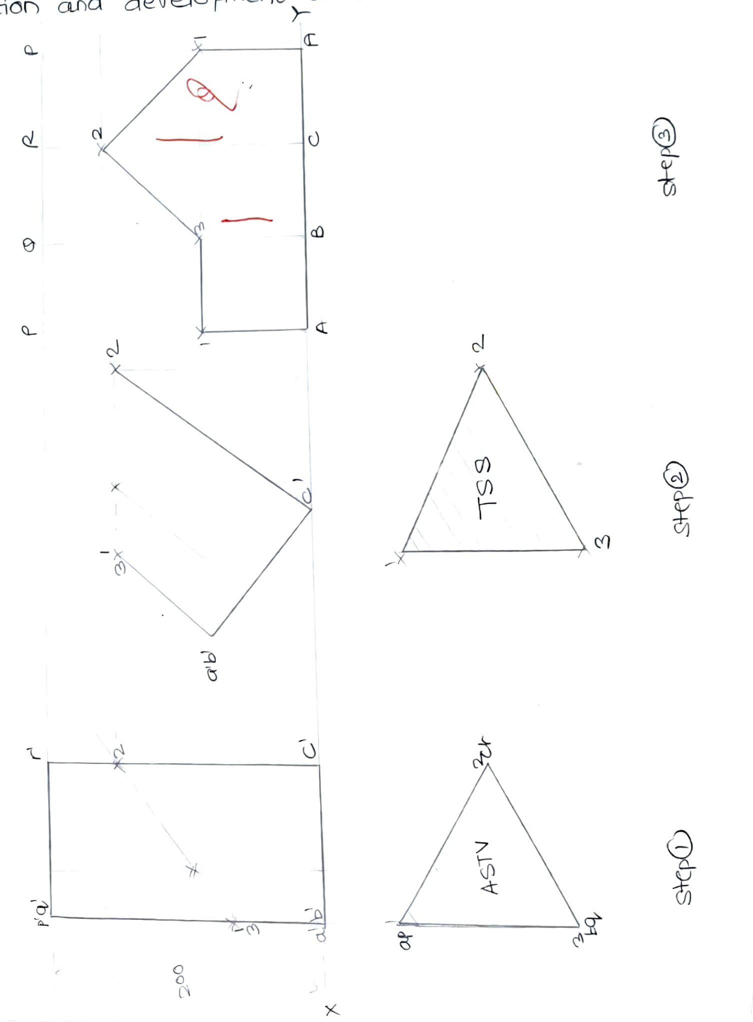
② Tetrahedron: 4 equilateral △



① Cylinder of 150mm diammeter and 200 mm height rests on HP on its circular base. It is cut by an AIP cutting plane passing through the midpoint of the axis and inclined 45° to HP. Draw projections showing sectional TV and True shape of the section and development of cut cylinder.

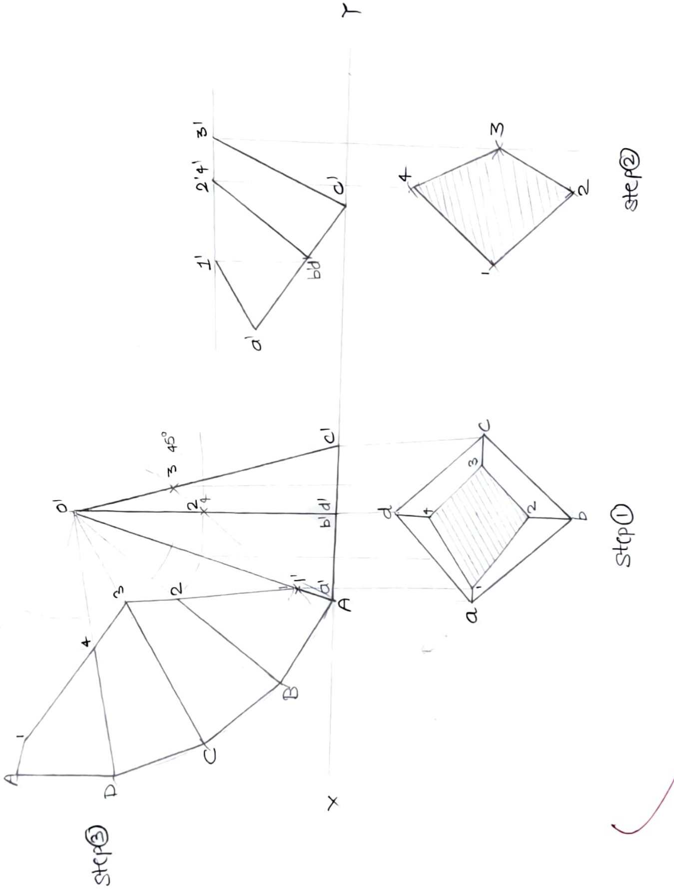


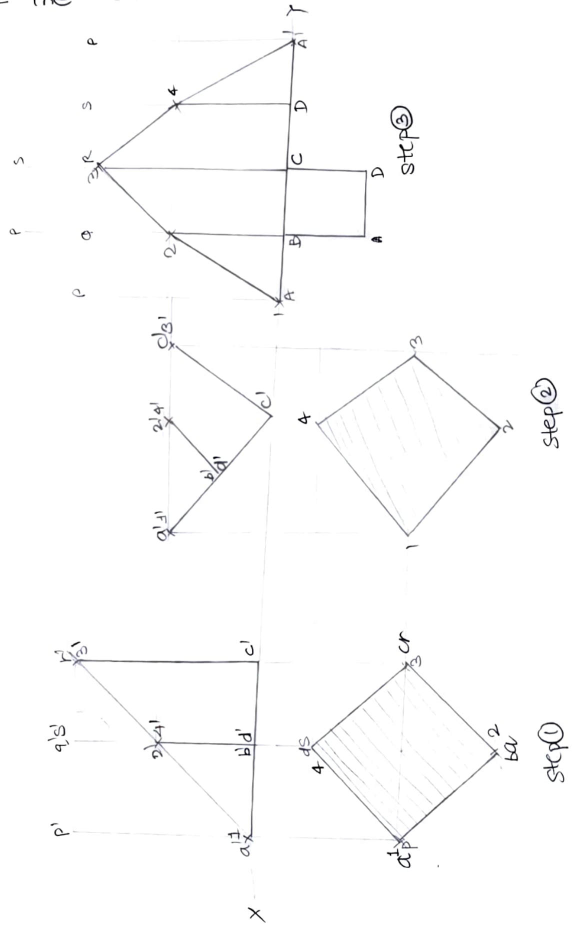
② Triangular prism base edge 150mm and height 200mm rests on its base on the HP, with one of the base edge perpendicular to VP. Its is cut by an AIP cutting section plane inclined 45° to base / HP, bisecting the axis. Draw projection showing sectional TV, true shape of the section and development of cut solid (prism)



(3) Square Pyramid of base edge 100mm and axis 200mm is resting on HP on its base such that base edges are equally inclined to VP. It is cut by an AIP section / cutting plane, bisecting the axis, inclined 45° to the HP. Draco projections shocoing sectional TV, True shape of the section and development of cut pyramid

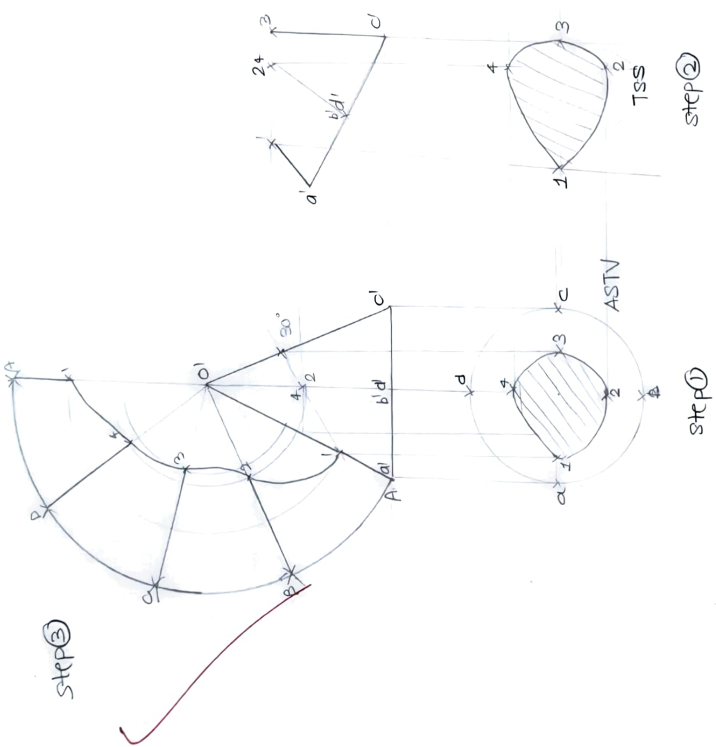
(4) Cube of 150mm long edges is resting on one of its faces on HP with four verticle faces equally inclined to VP. It is cut by an AIP cutting/section plane passing through the body diagonal of a cube. Draco projections shoeing location of cutting plane, sectional TV, True shape of the section and development of cut cube.

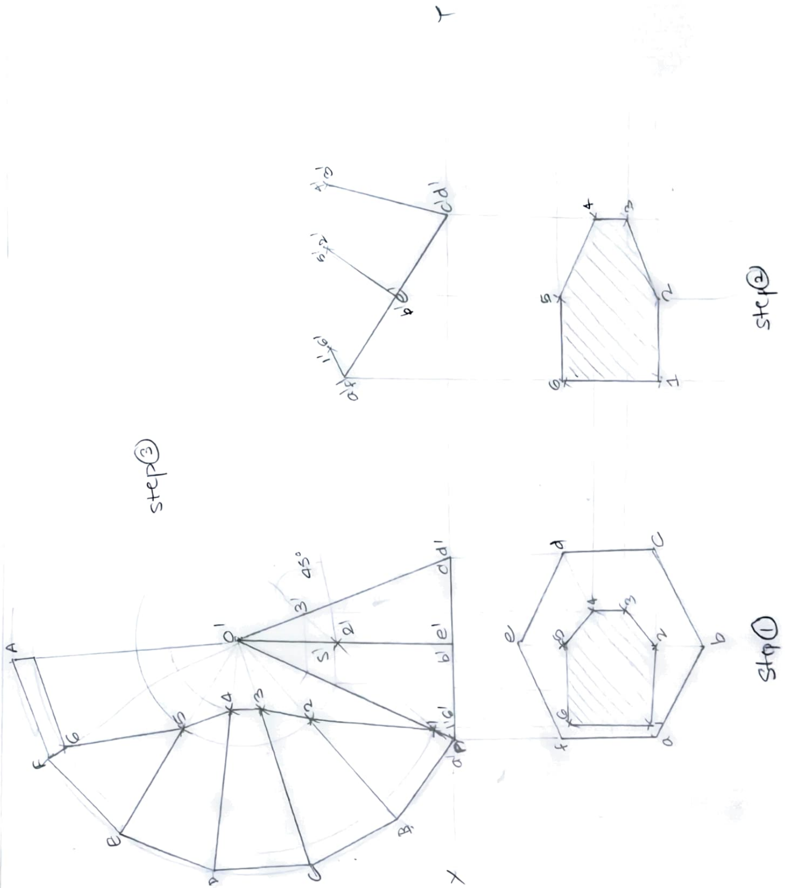




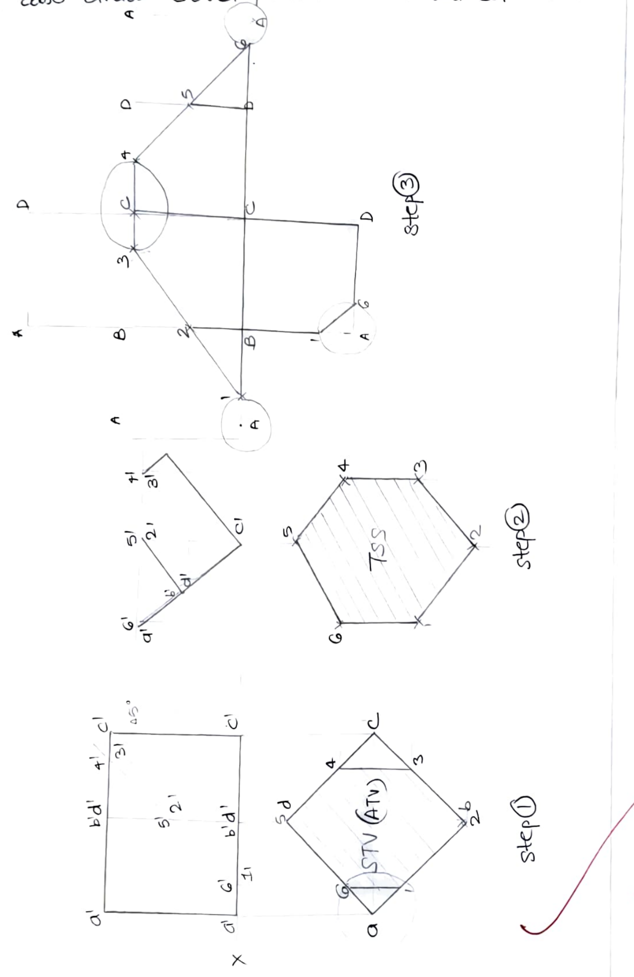
(5) Cone (diameter 150mm and height 125 mm) is standing on its base on the HP. It is cut by an AIP section/cutting plane, inclined 30° to the HP and passing through a point on axis 75mm from the base. Draw projections showing sectional TV, True shape of the section and development curved surface of cut cone.

(6) Hexagonal pyramid of base side 100mm and height 200 mm is resting on its base on HP, such that two base edges are perpendicular to VP. It is cut by an AIP section plane, inclined 45° to base and bisecting the axis. Draw sectional TV, true shape of the section and also draw development of retained solid.





7. Cube of 150mm long edge is resting on one of its face on HP with four verticle faces equally inclined to VP. It is cut by an AIP section plane inclined 45° to HP. Bisecting the axis. Draw sectional TV, true shape of the section and also draw development of retained solid.



V. good